

Model Comparison: Typo Effect Analysis

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$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

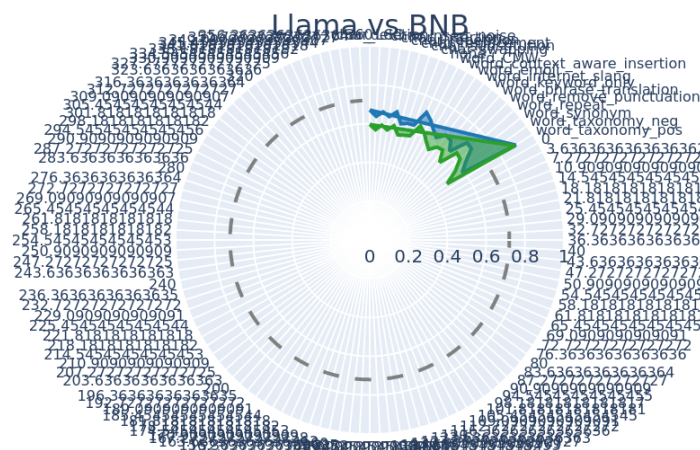
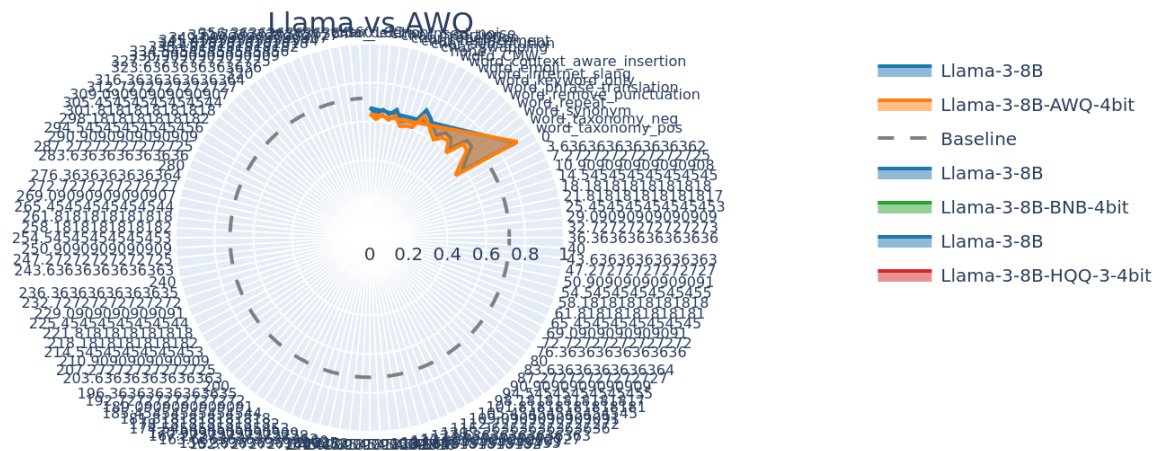
Configuration Details

batch_size: 32
dataset_name: P101, P103, P108, P127, P1376, P1412, P159, P17, P176, P178, P19, P20, P264, P27, P276, P30, P364, P37, P495, P740
exp_id: pert-awq-bnb-hqq-10-07
max_entries: None
max_new_tokens: 25
model_name: Llama-3-8B, Llama-3-8B-AWQ-4bit-local, Llama-3-8B-BNB-4bit-local, Llama-3-8B-HQQ-mixed-local
n_beams: 5
n_repeats: 1
strategy: Direct Completion
temperature: 0.1
typo_intensity: 1, 2, 3
typo_type: none, char_insertion, char_deletion, char_replacement, char_repetition, char_swapping, word_CMW, char_LCC, word_synonym, char_insert_noise, word_repeat, char_substitution, word_emoji, word_internet_slang, word_phrase_translation, word_context_aware_insertion, word_remove_punctuation, word_keyword_only, word_taxonomy_pos, word_taxonomy_neg
use_beam_search: False

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Comparison Radar Plot of Accuracy for Intensity 1

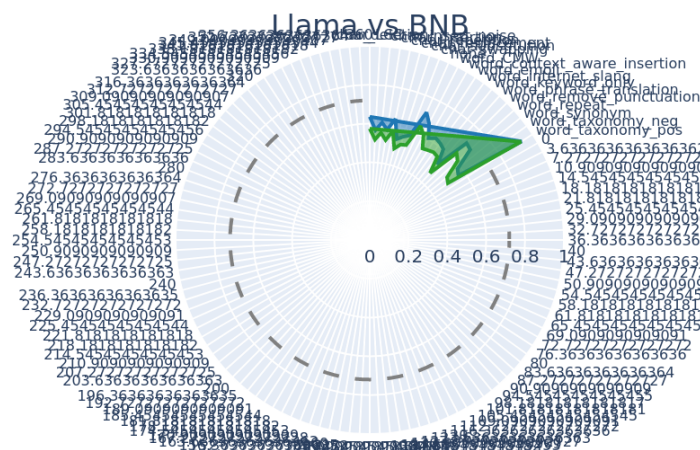
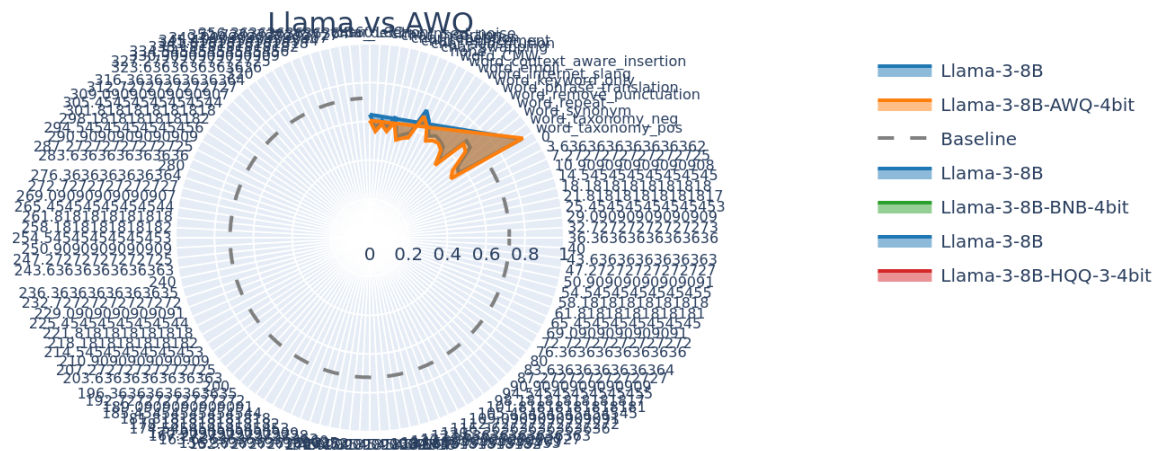
Comparison of Average Accuracy for Intensity 1 Across All Datasets



$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Comparison Radar Plot of Accuracy for Intensity 2

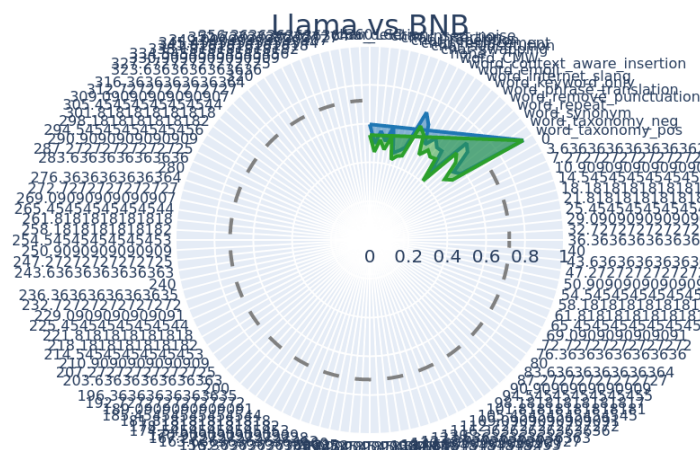
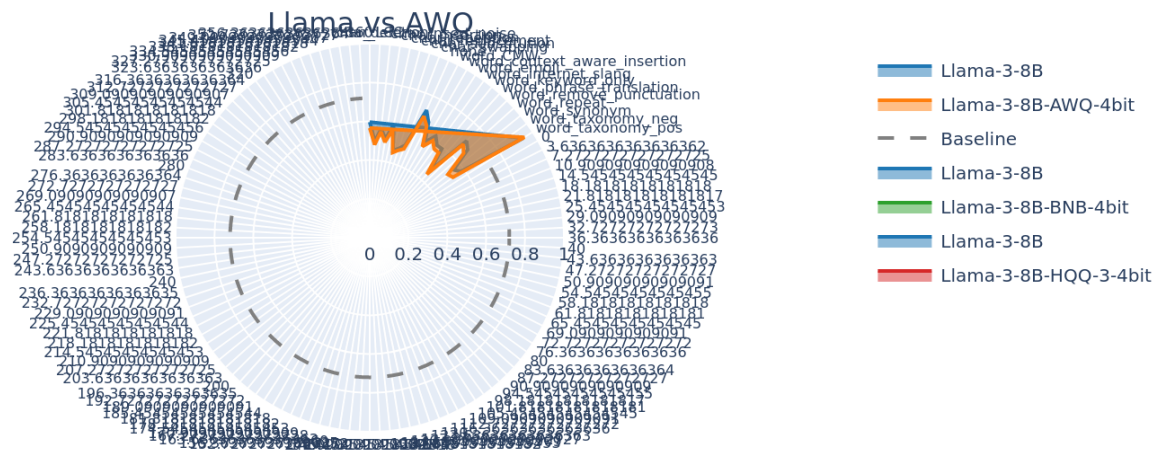
Comparison of Average Accuracy for Intensity 2 Across All Datasets



$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Comparison Radar Plot of Accuracy for Intensity 3

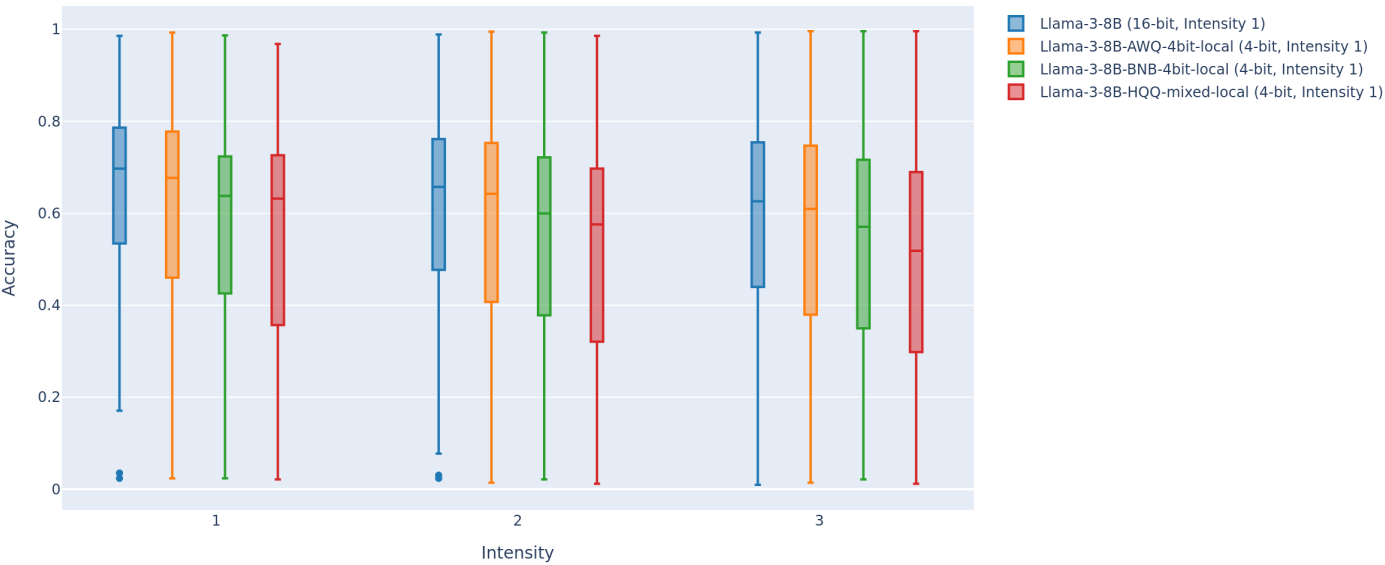
Comparison of Average Accuracy for Intensity 3 Across All Datasets



$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Boxplot Comparison of Accuracy

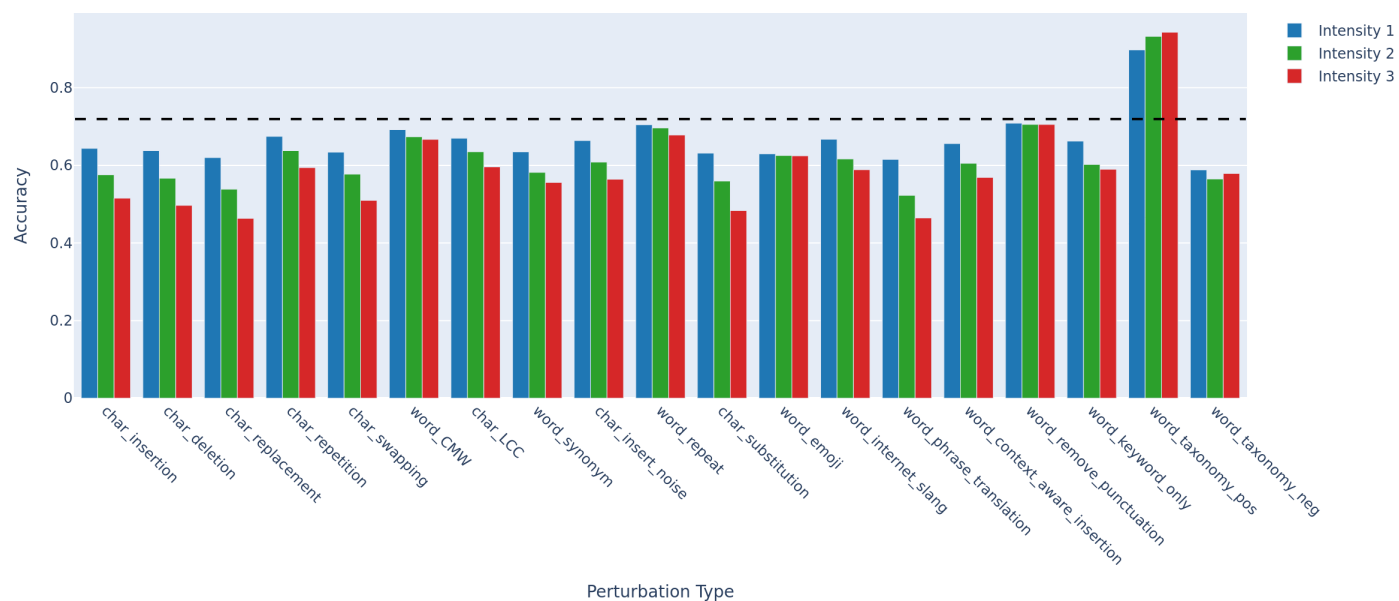
Distribution of Accuracy across Intensities and Models



$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

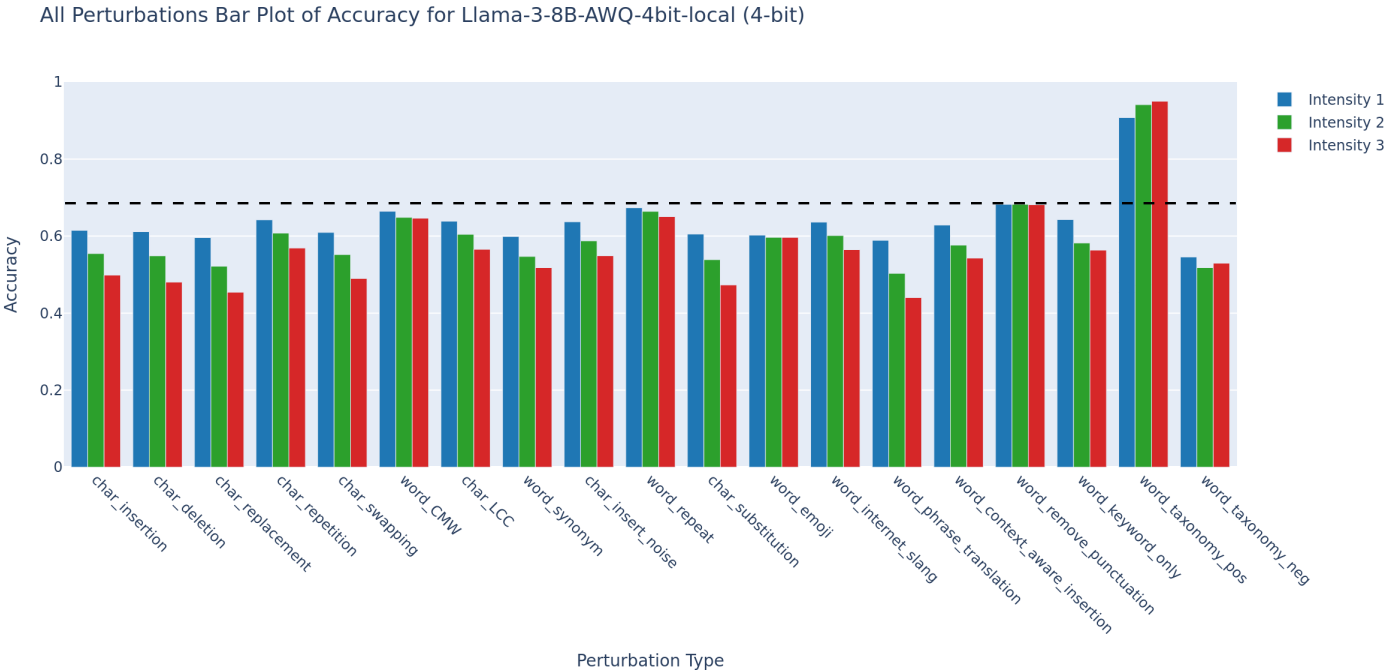
All Perturbations Bar Plot of Accuracy for Llama-3-8B

All Perturbations Bar Plot of Accuracy for Llama-3-8B (16-bit)



$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

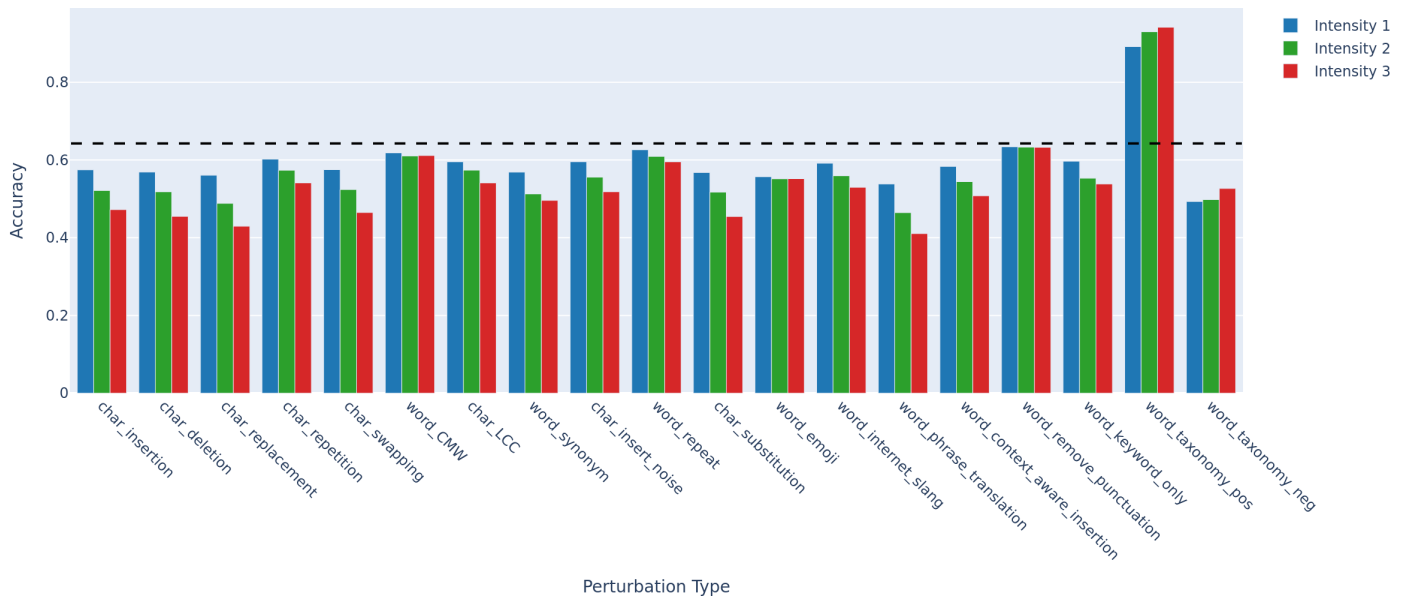
All Perturbations Bar Plot of Accuracy for Llama-3-8B-AWQ-4bit-local



$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

All Perturbations Bar Plot of Accuracy for Llama-3-8B-BNB-4bit-local

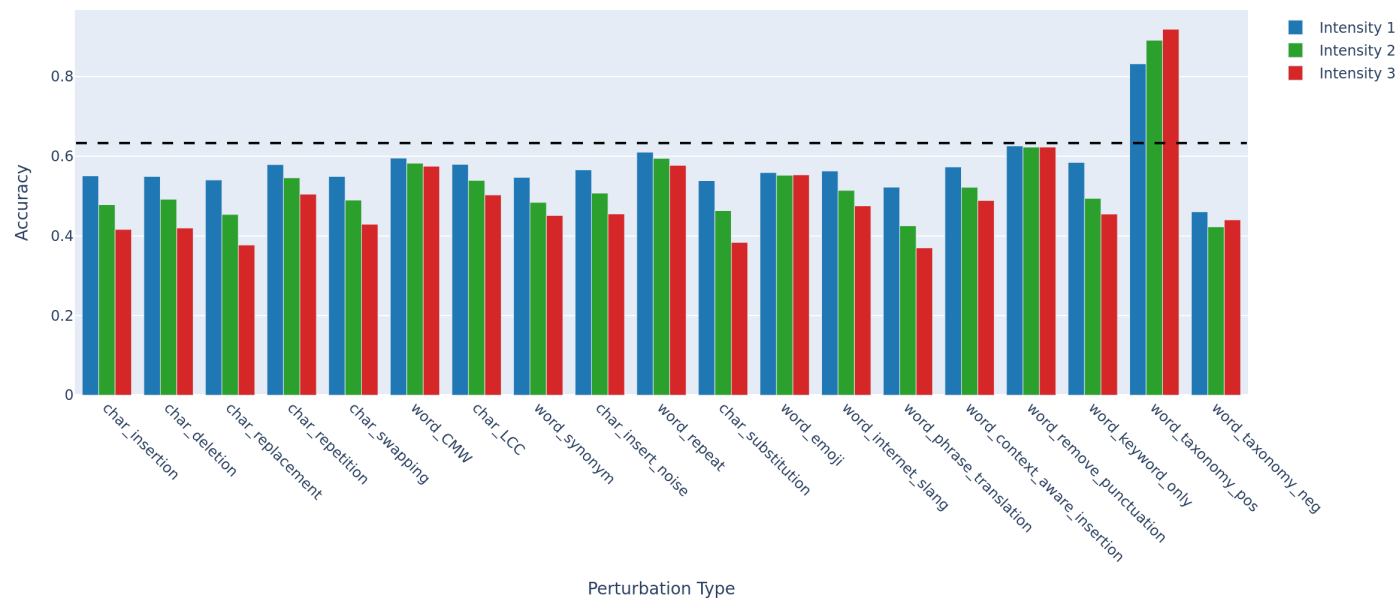
All Perturbations Bar Plot of Accuracy for Llama-3-8B-BNB-4bit-local (4-bit)



$$\text{AUCPR} = \int_0^1 \frac{TP}{TP+FP} d\left(\frac{TP}{TP+FN}\right)$$

All Perturbations Bar Plot of Accuracy for Llama-3-8B-HQQ-mixed-local

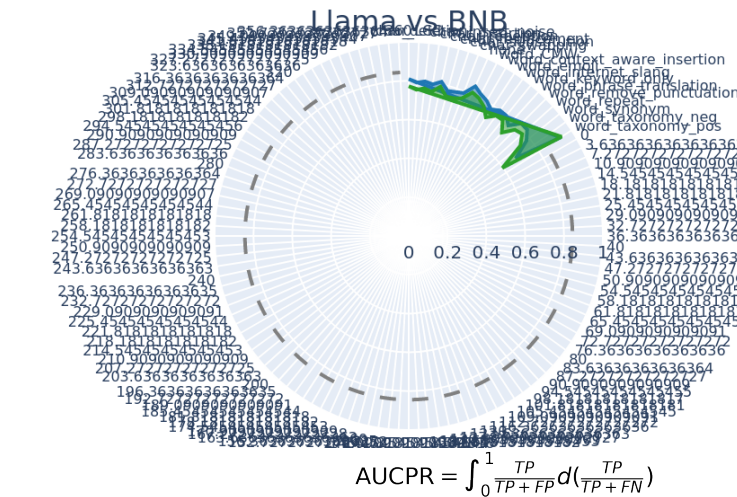
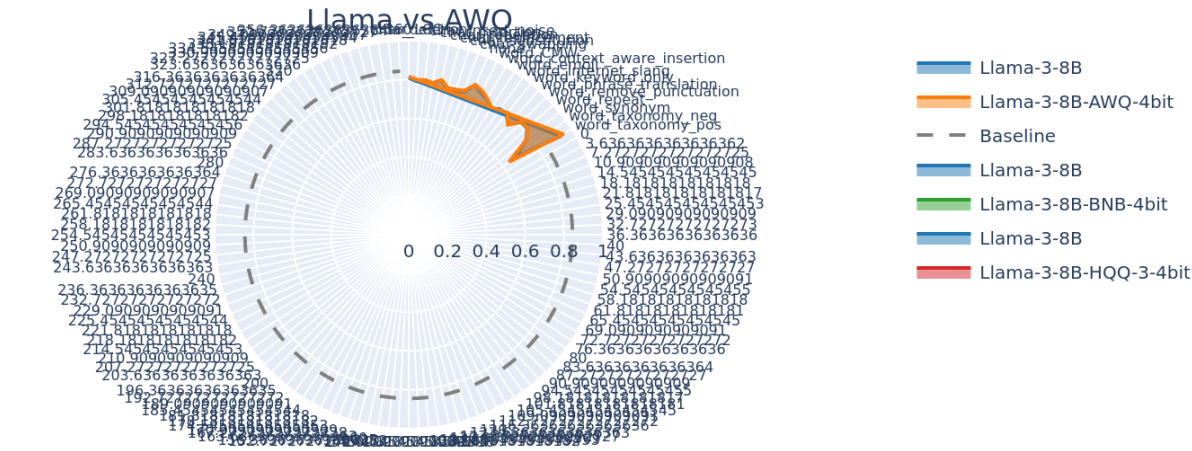
All Perturbations Bar Plot of Accuracy for Llama-3-8B-HQQ-mixed-local (4-bit)



$$\text{AUCPR} = \int_0^1 \frac{TP}{TP+FP} d\left(\frac{TP}{TP+FN}\right)$$

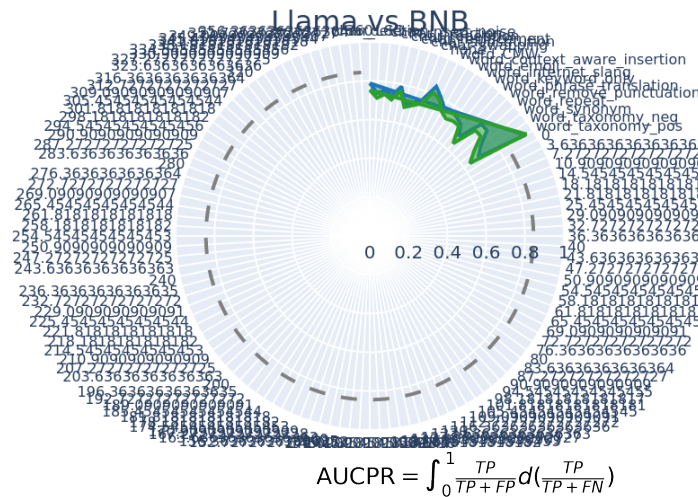
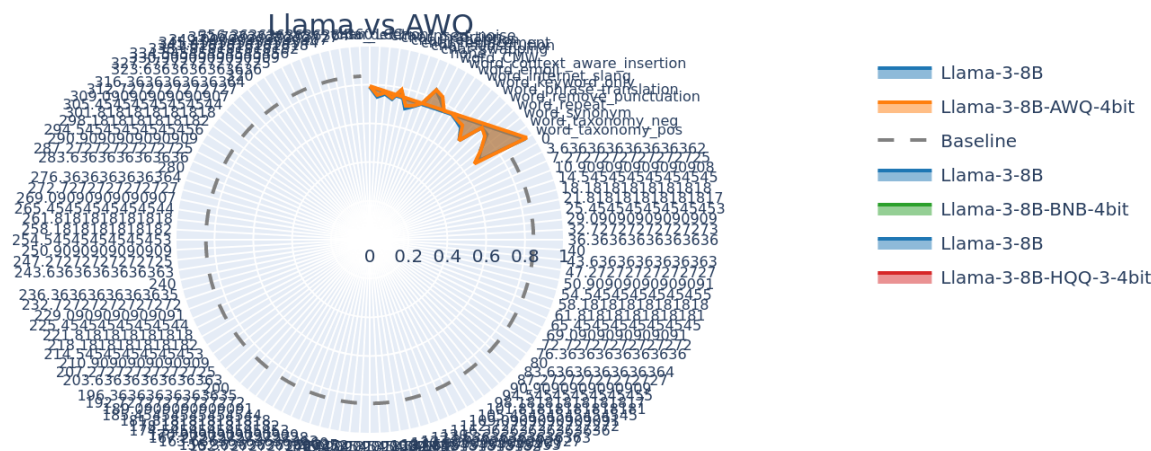
Comparison Radar Plot of AUCPR_sample for Intensity 1

Comparison of Average AUCPR_sample for Intensity 1 Across All Datasets



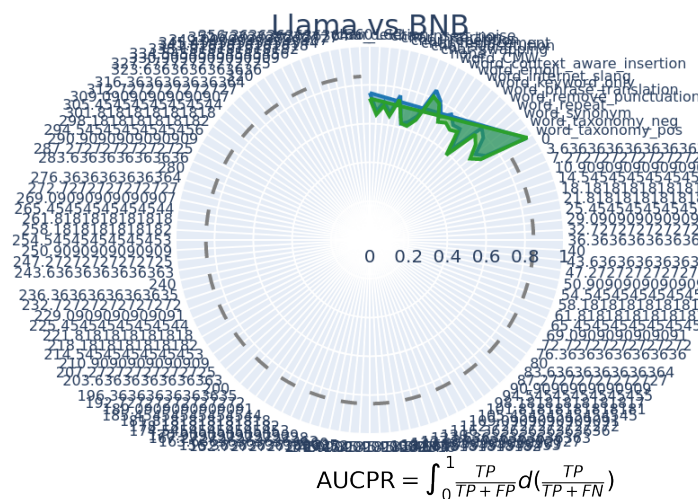
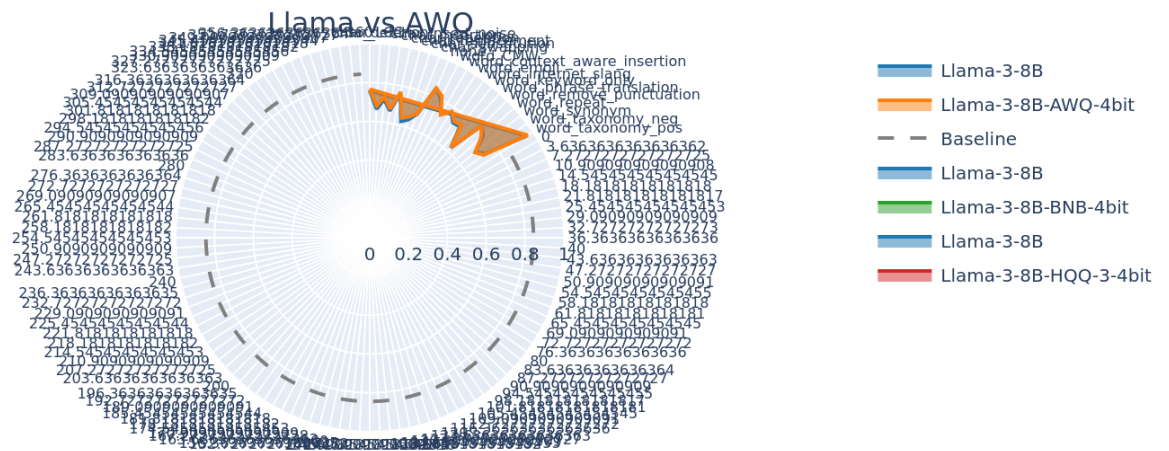
Comparison Radar Plot of AUCPR_sample for Intensity 2

Comparison of Average AUCPR_sample for Intensity 2 Across All Datasets



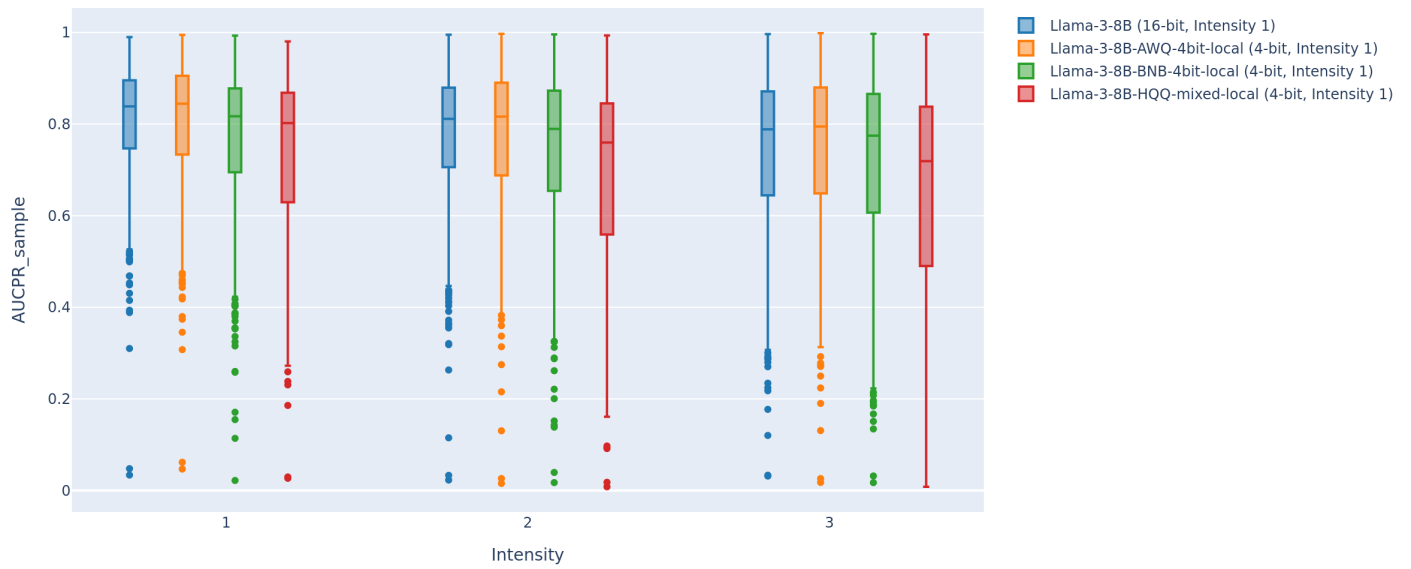
Comparison Radar Plot of AUCPR_sample for Intensity 3

Comparison of Average AUCPR_sample for Intensity 3 Across All Datasets



Boxplot Comparison of AUCPR_sample

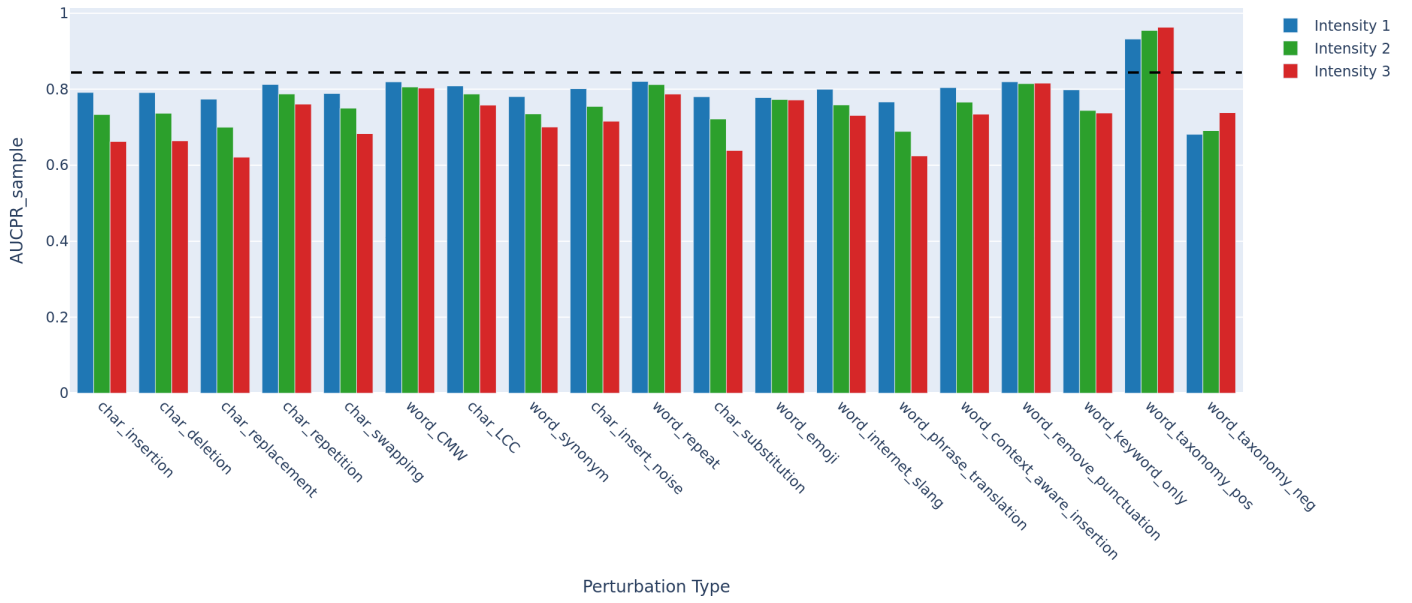
Distribution of AUCPR_sample across Intensities and Models



$$\text{AUCPR} = \int_0^1 \frac{TP}{TP+FP} d\left(\frac{TP}{TP+FN}\right)$$

All Perturbations Bar Plot of AUCPR_sample for Llama-3-8B

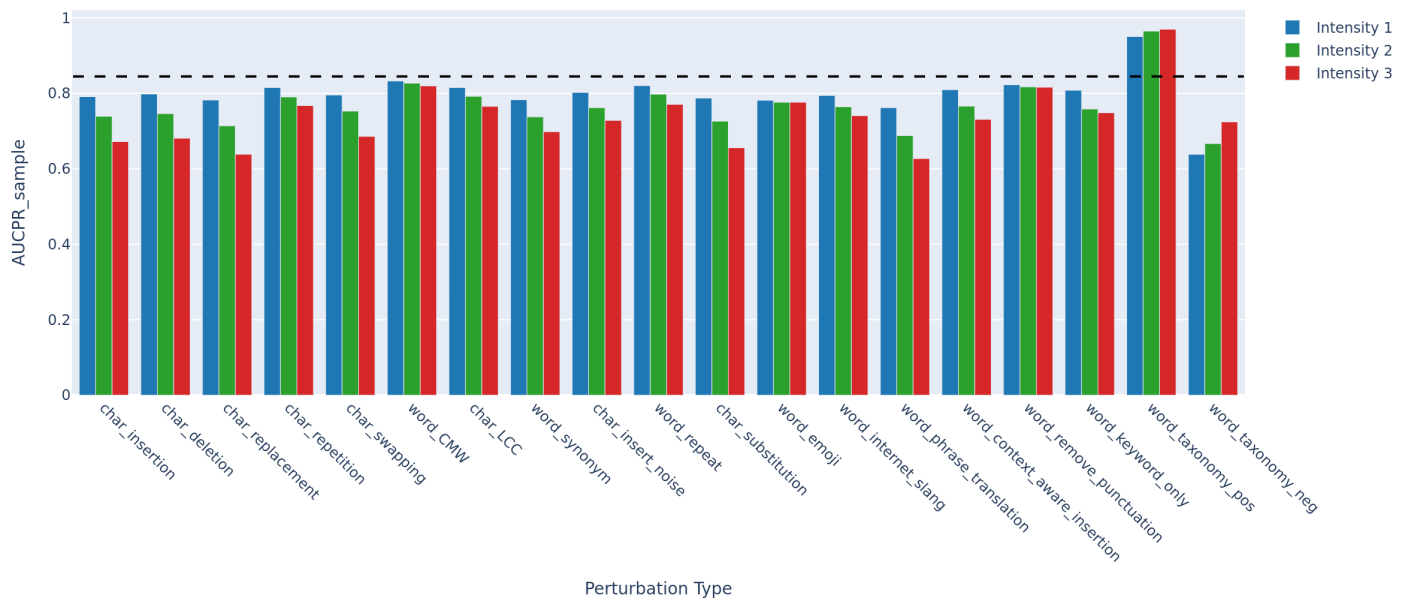
All Perturbations Bar Plot of AUCPR_sample for Llama-3-8B (16-bit)



$$\text{AUCPR} = \int_0^1 \frac{TP}{TP+FP} d\left(\frac{TP}{TP+FN}\right)$$

All Perturbations Bar Plot of AUCPR_sample for Llama-3-8B-AWQ-4bit-local

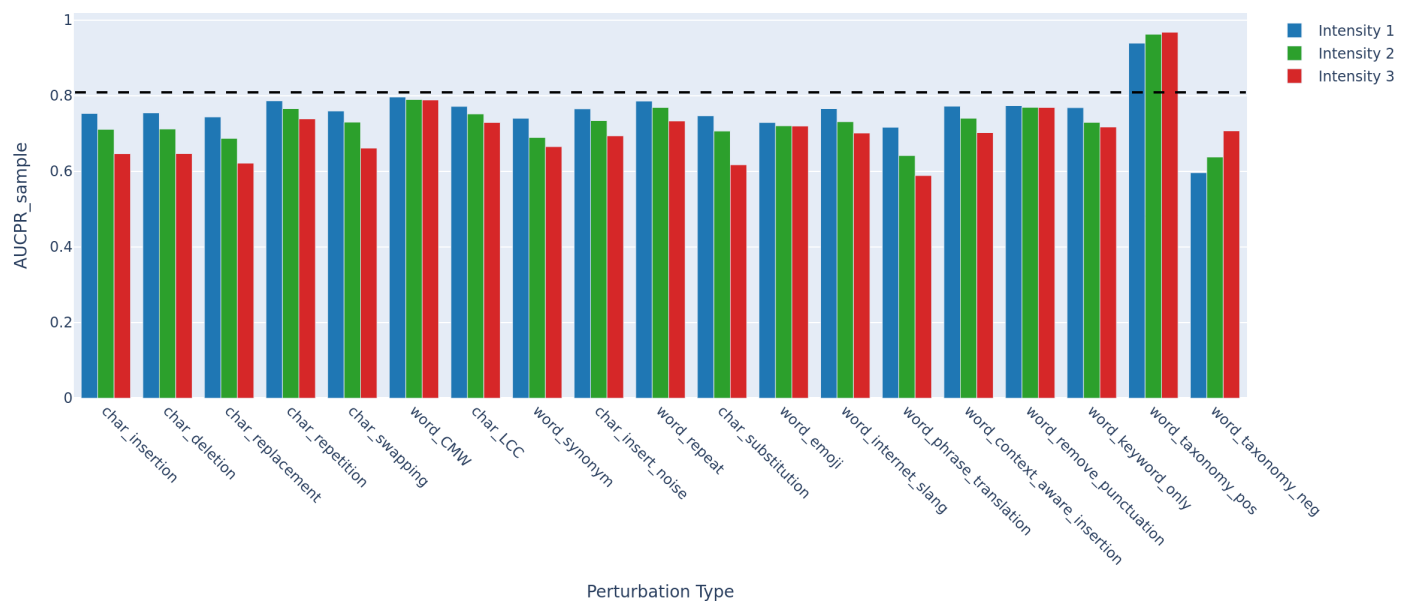
All Perturbations Bar Plot of AUCPR_sample for Llama-3-8B-AWQ-4bit-local (4-bit)



$$\text{AUCPR} = \int_0^1 \frac{TP}{TP+FP} d\left(\frac{TP}{TP+FN}\right)$$

All Perturbations Bar Plot of AUCPR_sample for Llama-3-8B-BNB-4bit-local

All Perturbations Bar Plot of AUCPR_sample for Llama-3-8B-BNB-4bit-local (4-bit)



All Perturbations Bar Plot of AUCPR_sample for Llama-3-8B-HQQ-mixed-local

